Seat 28

Stellenbosch UNIVERSITY IYUNIVESITHI UNIVERSITEIT

FACULTY OF ENGINEERING/ FAKULTEIT INGENIEURSWESE

Module Code: 18791-354

ENGINEERING ECONOMY 354 INGENIEURSEKONOMIE

ASSESSMENT A3/ ASSESSERING A3

CLOSED BOOK/ TOEBOEK

PAPER 1 OF 1 / VRAESTEL 1 VAN 1

DATE/ DATUM	November 2023
DURATION/ TYDSDUUR	2 hours / 2 ure
TOTAL OF MARKS/ PUNTE TOTAAL	109

Certification/ Sertifisering

	TITLE, NAME AND SURNAME/ TITEL, NAAM EN VAN	SIGNATURES CONFIRM: Question paper and marking scheme are correct and aligned with module outcomes. HANDTEKENINGE BEVESTIG: Vraestel en merk-skema is korrek en belyn met module uitkomste.									
Questions:		Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Total
Marked out of:		6	6	22	14	12	8	10	16	15	109
EXAMINER/ EKSAMINATOR:	**TG Dirkse-van Schalkwyk**	6	0	20	0	10	3	0	13½	12	64.5
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Signature/ Handtekening:	
Declaration By placing my signature in the space provided, I confirm: [1] that I have read, understand, and will abide by the assessment rules and instructions set out on the question paper and assessment answer book [2] that I am aware that a contravention of an assessment rule or instruction may cause an assessment to be declared null and void and may even lead to expulsion. [3] that the particulars furnished by me on this front cover are true and correct.	**Verklaring** Deur my handtekening in die onderstaande ruimte/s te plaas, bevestig ek: [1] dat ek die assesseringsreëls en voorskrifte op die vraestel van die antwoordboek gelees het, verstaan en sal gehoorsaam; [2] dat ek bewus is 'n oortreding van 'n assesseringsreël of voorskrif daartoe kan lei dat 'n assessering ongeldig verklaar word, en selfs tot skorsing kan lei; [3] dat die besonderhede wat ek op hierdie blad verstrek het, waar en korrek is.

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Special Instructions/ Instruksies

1. Read the information on both this question paper and the accompanying examination book, provide your details on both, and sign both.	1. Lees die inligting op beide hierdie vraestel en die meegaande eksamenboek, verskaf u gegewens op beide, en teken beide.
2. Make sure your student number and signature are on the front of ALL your answer book(s), before you hand in.	2. Voordat u inhandig, maak seker dat u studentenommer en handtekening op AL u antwoordboek(e) aangebring is.
3. The Engineering Faculty's General Stipulations for under- and postgraduate modules" apply to this assessment.	3. Fakulteit Ingenieurswese se "Algemene bepalings vir voor- en nagraadse modules" is op hierdie assessering van toepassing.
4. The question paper consists of 9 questions. Make sure that you have received TWENTY printed pages.	4. Die vraestel bestaan uit 9 vrae. Maak seker dat u TWINTIG gedrukte bladsye ontvang het.
5. This is a closed book assessment. One A4 page, written both sides may be brought in, with only formulas.	5. Hierdie is 'n toeboek assessering. Een A4 beide kante geskryf mag ingebring word, met slegs formules.
6. Pocket calculators prescribed for the first two years of the BEng may be used.	6. Sakrekenaars voorgeskryf vir die eerste twee jaar van die Bing mag gebruik word.
7. TABLES are not provided.	7. TABELLE word nie voorsien nie.
8. Answer all questions.	8. Beantwoord al die vrae.
9. Answer all the questions in the spaces provided on this question paper.	9. Beantwoord al die vrae in die ruimtes voorsien op hierdie vraestel.
10. If, in your opinion, there are any missing or incorrect parameters, make reasonable assumptions, state them clearly and continue.	10. As daar na jou mening enige parameters ontbreek of verkeerd is, maak redelike aannames, stel jou aannames duidelik en gaan voort.
11. The marks for each question are indicated to the right of the question in brackets.	11. Die punte vir elke vraag word regs van die vraag in hakies aangedui.
12. No part of the question/answer set may be torn out.	12. Geen deel van die vraag/antwoordstel mag uitgeskeur word nie.
13. ALL the pages of the question paper must be handed in.	13. AL die bladsye van die vraestel moet ingehandig word.
14. Put the completed question paper inside the rough-work book and hand both (everything) in.	14. Plaas die voltooide vraestel in die rofwerkboek en handig beide (volledig) in.
15. The accompanying book is for rough-work only and will not be marked.	15. Die meegaande antwoordboek is slegs vir rofwerk beskikbaar word nie gemerk

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	nie.
16. No use of electronic media in the venue.	16. Geen elektroniese media mag in die lokaal gebruik word nie.
17. Ensure that you are in the correct venue according to the subdivision of surnames. If not, go to the correct room.	17. Maak seker u sit in die regte lokaal volgens u van-indeling. Indien nie, gaan na die regte lokaal.

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QUESTION 1/ VRAAG 1 [6]

6

A government bond pays a dividend at a nominal rate of 20% per year, compounded monthly. The bondholders pay tax of 11.02% while inflation is 6.4% per year. What is the real rate they receive on their investment after taking tax and inflation respectively into account? Show all steps to get marks for the method.	'n Regerings-skuldbrief betaal 'n dividend teen 'n nominale koers van 20% per jaar, maandeliks saamgestel. Die verbandhouers betaal belasting van 11.02% terwyl inflasie 6.4% per jaar is. Wat is die reële rentekoers wat hulle ontvang op hulle belegging nadat hulle belasting en inflasie onderskeidelik in ag geneem het? Wys alle stappe om punte te kry vir die metode.

$i_{eff} = \left(1 + \frac{0,20}{12}\right)^{12} - 1 = 0,219391$

$i_{tax} = i_{eff}(1 - T) + 1$

$= 0,219391(1 - 0,1102) + 1$

$= 1,195214$

$i_{real} = \frac{i_{tax}}{1 + f} - 1 = \frac{1,195214}{1 + 0,064} - 1 = 0,12332160 5$

$= 12,332\%$ ✓

6

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QUESTION 2/ VRAAG 2 [6]

A company bought a machine with a useful life of 6 years for R5.7m. The salvage value is R1.6m. This machine may normally be depreciated over 6 years on a straight-line basis. Government however wants to stimulate the economy, and introduced an accelerated depreciation basis in three years, allowing depreciation of 50% in year one, and 30% in year two and 20% in year 3. 2.1 With the company paying 28% tax, and having a MARR of 35%, calculate the two alternatives' present worths using MARR to discount. Assume that the company makes about R15m per year profit. [5] 2.2 Which method is the best to use? [1] Show all steps to get marks for the method.	'n Maatskappy het 'n masjien met 'n nuttige lewensduur van 6 jaar vir R5,7 miljoen gekoop. Die bergingswaarde is R1.6m. Hierdie masjien kan normaalweg oor 6 jaar op 'n reguitlynbasis afgeskryf word. Die regering wil egter die ekonomie stimuleer, en het 'n versnelde waardeverminderingsbasis in drie jaar ingestel, wat waardevermindering van 50% in jaar een, en 30% in jaar twee en 20% in jaar 3 toelaat. 2.1 Met die maatskappy wat 28% belasting betaal, en met 'n MARR van 35%, bereken die huidige waardes van die twee alternatiewe deur met MARR te verdiskonteer. Neem aan dat die maatskappy ongeveer R15m per jaar wins maak. [5] 2.2 Watter metode is die beste om te gebruik? [1] Wys alle stappe om punte te kry vir die metode.

1 ~~Old PW:~~

~~Depreciation = $\frac{5,7}{6}$ = R0,95 m / year~~

~~Tax savings = Depreciation x Tax~~
~~= 0,95 x 0,28) = R0,266 m / year~~

PW

1

Year	Straight line	Accelerated
0	$-5,7 + 15 - 0,28(15)$	
1	$P - T(P - \frac{D}{M})$	
2		
3		

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QUESTION 3/ VRAAG 3 [22]

A project requires an initial investment of R12m for hardware and R2.1m for software at year 0. The project will require additional amounts linked to the software inflation rate, to upgrade the software after every 3 years. Currently, a software update is valued at R1060k and software inflation is expected to be 11.3%. The project life is five years and the salvage value for the hardware is estimated at R3m. The hardware may be depreciated using an accelerated method with 30% in year one, 20% in year two, 20% in year three and 15% in years four and five. Software may be seen as a direct expense. The project income is estimated to be R7.5m in year one and will grow year on year by 19%. Operating costs will start at R1m in year 1 and increase by 8.1% per year. The company uses MARR of 31%, and has a tax rate of 28%.

a) What are the actual cash flows per year before tax for the project? [6]
b) What are the actual tax cash flows (payment or tax savings) for the project? Assume that the company makes about R10m profit from other projects. [6]
c) Calculate the net present value of the project when including tax? Use MARR to discount. [4]
d) Calculate the discounted payback period for the project using after tax cash flows. [4]
e) Explain why you may treat any loss from this project as a "tax saving" for the company? [2]

'n Projek vereis 'n aanvanklike belegging van R12m vir hardeware en R2.1m vir sagteware op jaar 0. Die projek sal addisionele bedrae vereis gekoppel aan die sagteware-inflasiekoers, om die sagteware elke 3 jaar op te gradeer. Tans word 'n sagteware-opdatering gewaardeer teen R1060k en sagteware-inflasie sal na verwagting 11,3% wees. Die projekleeftyd is vyf jaar en die skrootwaarde vir die hardeware word op R3m geraam. Die hardeware kan met 'n versnelde metode afgeskryf word met 30% in jaar een, 20% in jaar twee, 20% in jaar drie en 15% in jaar vier en vyf. Sagteware kan as 'n direkte uitgawe gesien word. Die projekinkomste word geraam op R7.5m in jaar een en sal jaar op jaar met 19% groei. Bedryfskoste sal in jaar 1 by R1 miljoen begin en met 8,1% per jaar styg. Die maatskappy gebruik MARR van 31%, en het 'n belastingkoers van 28%.

a) Wat is die werklike kontantvloei per jaar voor belasting vir die projek? [6]
b) 'Wat is die werklike belastingkontantvloei (betalings of belasting spaar) vir die projek? Aanvaar dat die maatskappy maak 10 miljoen wins uit ander projekte. [6]
c) Bereken die netto huidige waarde van die projek wanneer belasting ingesluit word? Gebruik MARR om te verdiskonteer. [4]
d) 'Bereken die verdiskonteerde terugbetalingstydperk vir die projek met behulp van nabelaste kontantvloei. [4]
e) Verduidelik hoekom jy enige verlies van hierdie projek as 'n "belastingbesparing" vir die maatskappy kan beskou? [2]

a)

Year	Cash flow before tax (mil)
0	$-14,1$ = R14,1 m
1	$7,5 - 1$ = R6,5 m
2	$7,5(1,19)^1 - 1(1,081)^1$ = R7,844 m
3	$7,5(1,19)^2 - 1(1,081)^2 - 1,06(1,113)^3$ = R7,9907 m
4	$7,5(1,19)^3 - 1(1,081)^3$ = R11,3755 m
5	$7,5(1,19)^4 - 1(1,081)^4 + 3$ = R16,6745 m

6

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b) initial investment: 12m

C-S = 9m

Year	Actual Tax Cash Flows
* 0	$-[-14,1+12) \times 0,28 = 0,588$
1	$-(6,5-0,3(9)) \times 0,28 = -1,064$
2	$-(7,844-0,2(9)) \times 0,28 = -1,69232$
3	$-(7,9907-0,2(9)) \times 0,28 = -1,7334$
4	$-(11,3755-0,15(9)) \times 0,28 = -2,8071$
5	$-(16,6745-9-0,15(9)) \times 0,28 = -1,7709$

5

c)

Year	NPV	Accum
0	$(-14,1+0,588)/(1,31)^0 = -13,512$	-13,512
1	$(6,5-1,064)/(1,31)^1 = 4,1496$	-9,3624
2	$(7,844-1,69232)/(1,31)^2 = 3,5847$	-5,7777
3	$(7,9907-1,7334)/(1,31)^3 = 2,7834$	-2,9943
4	$(11,3755-2,8071)/(1,31)^4 = 2,9095$	-0,0848
5	$(16,6745-1,7709)/(1,31)^5 = 3,8631$	3,7783
	NPV = R3,7783m	

Marked with error

4

d)

$$\frac{y-4}{5-4} = \frac{0-(-0,0848)}{3,7783-(-0,0848)}$$

$y = 4,000002244 \approx 4$ years

4

e) Because ↑ in profit means ↑ in tax, hence costing more. Loss is risk but less tax

1

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QUESTION 4/ VRAAG 4 [14]

0

Four mutually exclusive projects are listed below. All four options are expected to boost annual revenues by varying amounts. The rows below include all the estimates for the four options. Using MARR = 19% and inflation = 5%, perform PW evaluations for each project to select the option that is the most promising economically. What is the present value of the option that is most profitable? Show all the values in the columns for full marks.	Vier onderling uitsluitende projekte word hieronder gelys. Al vier opsies sal na verwagting jaarlikse inkomste met wisselende bedrae verhoog. Die rye hieronder sluit al die skattings vir die vier opsies in. Gebruik MARR = 19% en inflasie = 5%, voer PW-evaluasies vir elke projek uit om die opsie te kies wat ekonomies die belowendste is. Wat is die huidige waarde van die opsie wat die mees winsgewend is? Wys al die waardes in die kolomme vir volpunte.

Alternative	A	B	C	D
Initial cost	-R24k	-R28k	-R36k	-R48k
Annual income	R8k	R12k	R13k	R14k
Salvage value	R2k	R4k	R6k	R8k
Life, years	2	3	6	6

Fill the table below with your values (In the right-most column show which two projects you will compare example **Incr CF(B - A)**

Year	CashFlow A	CashFlow B	CashFlow C	CashFlow D	Incr CF(-)
0					
1					
2					
3					
4					
5					
6					
PW(5%)					
PW(MARR)					
Selected? Y/N					

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QUESTION 5/ VRAAG 5 [12]

Given the following cash flow:	Gegee die volgende kontantvloei:
Year 0: -88000, Year 1: 18000, Year 2: 20000, Year 3: 22000, Year 4: 24000, Year 5: 26000	
a) Calculate the present value with an inflation rate of 0%. [4] b) Iterate three times to find an estimate for the IRR of the cash flow. Record your present value calculation with each iteration. [8]	a) Bereken die huidige waarde met 'n inflasiekoers van 0%. [4] b) Itereer drie keer om a skatting te kry vir die IRR van die kontantvloei. Boekstaaf jou huidige waarde met elke iterasie. [8]

4/7

a) rate — present value

rate	present value
0%	R22000

b)

rate	present value
5%	6407,6
10%	-6042,23
7,5734%	

Yr	PV(7,5734%)	Accum
0	-88000/(1,07573)^0 = -88000	-88000
1	18000/(1,07573)^1 = 16732,77	-71267,23
2	20000/(1,07573)^2 = 17285,05	-53984,18
3	22000/(1,07573)^3 = 17672,92	-36311,26
4	24000/(1,07573)^4 = 17927,24	-18589,02
5	26000/(1,07573)^5 = 18248,85	-340,17

6

Estimated IRR = 0,07443639 = 7,443639 %

year	PV (10%)	Accum
0	-88000/(1,1)^0 = -88000	-88000
1	18000/(1,1)^1 = 16363,64	-71636,36
2	20000/(1,1)^2 = 16528,93	-55107,43
3	22000/(1,1)^3 = 16528,93	-38578,5
4	24000/(1,1)^4 = 16392,32	-22186,18
5	26000/(1,1)^5 = 16143,95	-6042,23

Yr	PV(5%)	Accum
0	-88000/(1,05)^0 = -88000	-88000
1	18000/(1,05)^1 = 17142,86	-70857,14
2	20000/(1,05)^2 = 18140,59	-52716,55
3	22000/(1,05)^3 = 19004,43	-33712,12
4	24000/(1,05)^4 = 19704,86	-13964,26
5	26000/(1,05)^5 = 20371,68	6407,6

-6042,23 6407,6

$\therefore \frac{IRR_1 - 0,05}{0,1-0,05} = \frac{0-6407,6}{-6042,23-6407,6}$ $\frac{IRR_2 - 0,05}{0,07534-0,05} = \frac{0-6407,6}{-340,17-6407,6}$

$IRR_1 = 0,075734$ $IRR_2 = 0,074436$

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QUESTION 6/ VRAAG 6 [8]

a) Explain the logic of depreciation? [2] b) Explain what savings may realise from depreciation? [2] c) In what statements and what positions in those statements will depreciation be shown or make an impact? [2] d) Why would a company prefer to have larger depreciation earlier in the life of an asset? [2]	a) Verduidelik die logika van waardevermindering? [2] b) Verduidelik watter besparings kan realiseer deur waardevermindering ? [2] c) In watter state en watter posisies in daardie state sal depresiasie getoon word of 'n impak maak? [2] d) Waarom sou 'n maatskappy verkies om groter waardevermindering vroeër in die leeftyd van 'n bate te hê? [2]

a) Depreciation is essentially a decrease in value of an asset due to certain factors causing the quality to decrease of time

b) Tax savings as depreciation is not included in tax costs

c) Present worth of an asset when calculating tax cost/savings

d) In the longer term, more tax will be saved if depreciation is larger earlier in an assets life

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QUESTION 7/ VRAAG 7 [10]

A 3-year-old backup power system is being considered for early replacement. Its current market value is R20,000. Estimated future market values of the system are for years 1 to 5: R10 000, R8 000, R6 000, R2 000, and zero in year 5 and annual operating costs for years 1 to 5: R-5 000, R-6 500, R-8 000, R-9 500 and R-12 500 in year 5. (The minus shows these are costs.) What is the annual worth of having **a service life of another three years** if the interest rate is 10% per year? [10]	'n Drie jaar oue rugsteunkragstelsel word oorweeg vir vroeë vervanging. Sy huidige markwaarde is R20 000. Geskatte toekomstige markwaardes van die stelsel is vir jare 1 tot 5: R10 000, R8 000, R6 000, R2 000, en nul in jaar 5 en jaarlikse bedryfskoste vir jare 1 tot 5: R-5 000, R-6 500, R-8 000, R-9 500 en R-12 500 in jaar 5. (Die minus wys dit is kostes.) Wat is die jaarlikse waarde daarvan **om 'n dienslewe van nog drie jaar** te hê as die rentekoers 10% per jaar is? [10]

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13½/14

QUESTION 8/ VRAAG 8 [12]

A massive infrastucture project with CV values and expected cash flows for each year is shown below. The initial cost of the project is **R80m** and the useful life of the project is indefinite. Inflation is 8%. Given: CV = Standard Deviation / Mean	'n Massiewe infrastruktuurprojek met CV-waardes en verwagte kontantvloeie vir elke jaar word hieronder getoon. Die aanvanklike koste van die projek is **R80 miljoen en** die nuttige lewensduur van die projek is onbepaald. Inflasie is 8%. Gegee: CV = Standaardafwyking / Gemiddeld
The project has a CV of 0.11 and expected profit per year of R36m per year. Ignore possible salvage values and taxation.	Die projek het 'n CV van 0,11 en verwagte wins per jaar van R36m per jaar. Ignoreer moontlike bergingswaardes en belasting.

Normal Distribution probabilities

Z waardes :	**-2.0**	-1.9	-1.8	-1.7	-1.6	**-1.5**	-1.4	-1.3	-1.2	-1.1	**-1.0**
p(-inf:Z) :	**0.0228**	0.0287	0.0359	0.0446	0.0548	**0.0668**	0.0808	0.0968	0.1151	0.1357	**0.1587**

8.1 What is the expected net present value of the project? [4]	8.1 Wat is die verwagte netto huidige waarde van die projek? [4]

$PW = -C + \frac{P}{r} = -80 + \frac{36}{0,08} = R370\ m$ ✓

4

8.2 What net present value for the project can you be 95.54% certain of getting at least that? [4]	8.2 Watter netto huidige waarde vir die projek kan jy 95.54% seker wees dat jy ten minste dit sal kry? [4]

$\mu = 36.$ $1 - 0,9554 = 0,0446$ $\therefore$ $z = -1,7$

$CV = 0,11$

$CV = \frac{\sigma}{\mu}$ $\therefore$ $\sigma = CV \times \mu = 0,11 \times 36 = 3,96$

$z = \frac{x - \mu}{\sigma}$ $\therefore$ $x = z \times \sigma + \mu = -1,7 \times 3,96 + 36$

$x = 29,268$

$\therefore$ Certain $NPV = -80 + \frac{29,268}{0,08} = R285,85\ m$ ✓

4

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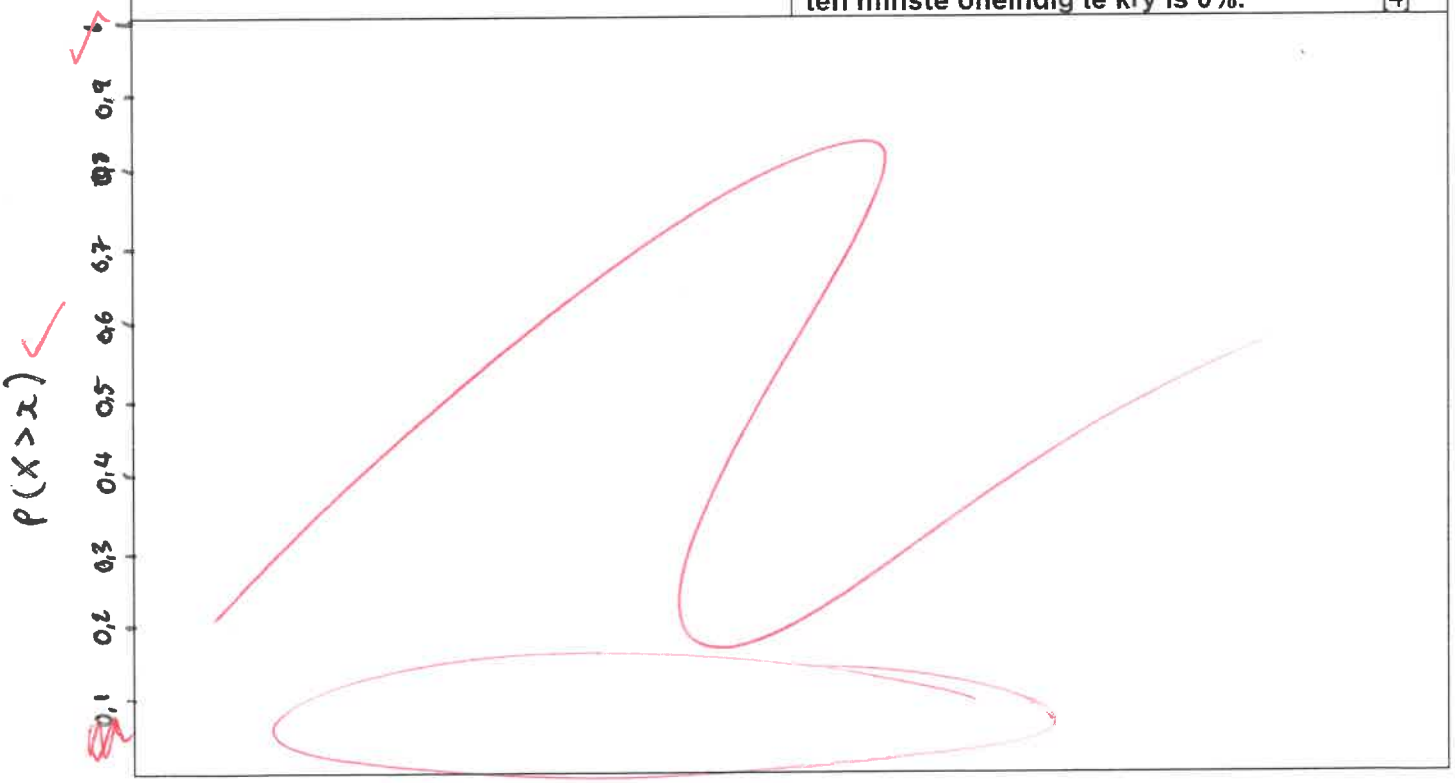
8.3 What is the discounted payback period for the project if the expected profit is made? [4]	8.3 Wat is die verdiskonteerde terugbetalings tydperk vir die projek indien die verwagte wins gemaak word? [4]

Years	PV	Accum
0	$-80/(1{,}08)^0 = -80$	-80
1	$36/(1{,}08)^1 = 33{,}3\dot{3}$	-46,6667
2	$36/(1{,}08)^2 = 30{,}8642$	-15,8025
3	$36/(1{,}08)^3 = 28{,}5780$	12,7755

$\therefore \frac{y-2}{3-2} = \frac{0-(-15{,}8025)}{12{,}7755-(-15{,}8025)} \quad \therefore y = 2{,}55296$ years

4

8.4 Draw an approximate graph that shows the probabilities (on the y-axis) of getting at least the potensial outcomes of the PW (on the x-axis) of the project. For instance the probability of getting less than -80m is zero and thus the **probability of getting a PW of at least -80m is 100%**. Similarly, the probability of getting less than infinity is 100% and the **probability of getting a PW of at least infinity is 0%.** [4]	8.4 Teken 'n benaderde grafiek wat die waarskynlikhede (op die y-as) toon om ten minste die potensiële uitkomste van die PW (op die x-as) van die projek te kry. Byvoorbeeld, die waarskynlikheid om minder as -80m te kry is nul en dus is die **waarskynlikheid om 'n PW van minstens -80m te kry 100%**. Net so is die waarskynlikheid om minder as oneindig te kry 100% en **die waarskynlikheid om 'n PW van ten minste oneindig te kry is 0%.** [4]



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12

QUESTION 9/ VRAAG 9 [15]

Given the following information for a business: regarding their cost of capital.

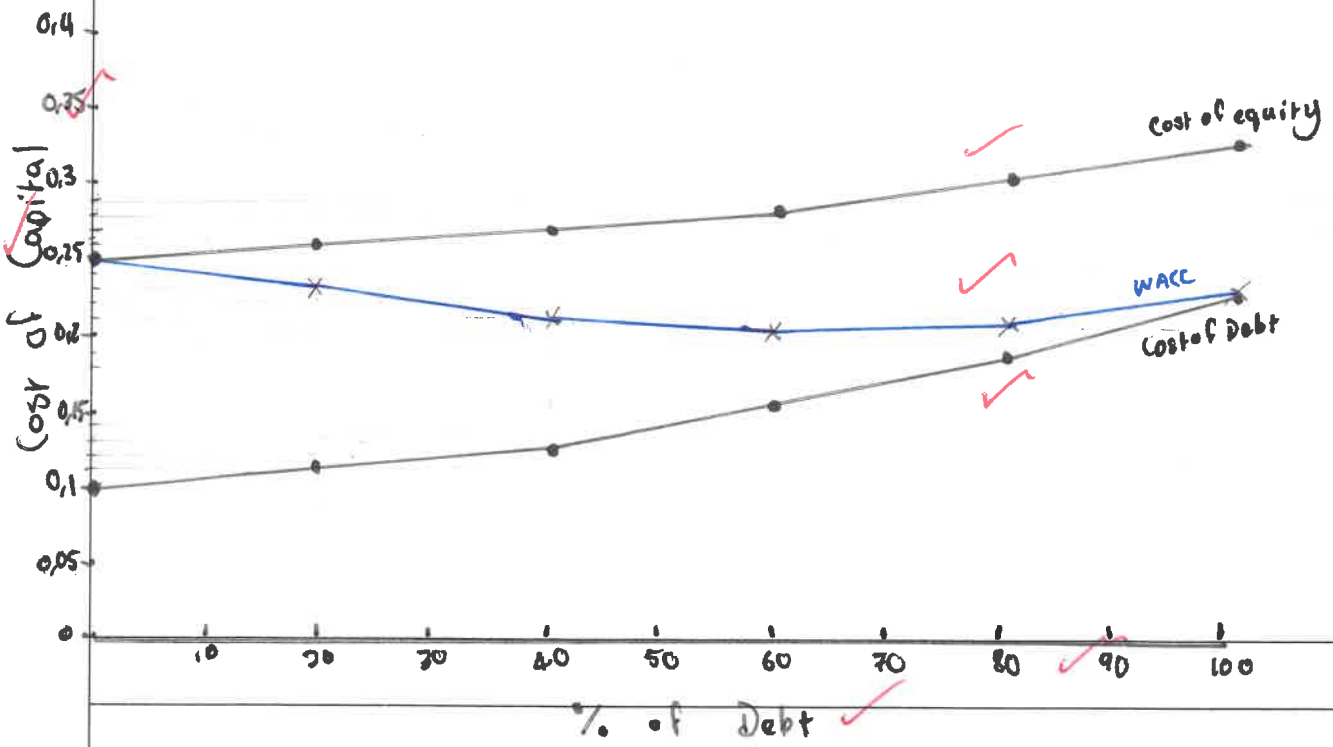
Gegewe die volgende informasie vir 'n besigheid rakende hulle koste van kapitaal:

Cost of Equity(Ekwiteit) = 25% and cost of Debt(Skuld) = 10% at 100% equity & 0% debt,
Cost of Equity = 26% and cost of Debt = 11% at 80% equity & 20% debt,
Cost of Equity = 27% and cost of Debt = 12% at 60% equity & 40% debt,
Cost of Equity = 28% and cost of Debt = 15% at 40% equity & 60% debt,
Cost of Equity = 30% and cost of Debt = 18% at 20% equity & 80% debt,
Cost of Equity = 32% and cost of Debt = 22% at 0% equity & 100% debt,

a) Draw the Cost of Equity, Cost of Debt and WACC on a graph with x-values as percentage Debt. [5]
b) If the business is using R180 000 of their own money, expecting a return of 27% on their equity, how much capital can they raise from a loan? [3]
c) Calculate the WACC for b) [2]
d) What is the lowest WACC for the business? [3]
e) What is the percentage Debt for the business in d) [2]

a) Teken die Koste van Equity, koste van Debt en WACC op 'n grafiek met die x-waardes as percentasie Debt. [5]
b) Indien die besigheid R180 000 van hulle eie geld gebruik en 'n opbrengs van 27% verwag op eie kapitaal, hoeveel kapitaal kan hulle kry vanuit 'n lening? [3]
c) Bereken die WACC vir b) [2]
d) Wat is die laagste WACC vir die besigheid? [3]
e) Wat is die persentasie Skuld vir d) [2]

a)



5.

b) 180000 x ((0,27 x 60%) + (0,12 x 40%)) = R 37800

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c) WACC = (0,27 x 60%) + (0,12 x 40%)
= 0,21 ✓3

d) $WACC_0$ = (0,25 x 100%) + (0,1 x 0%) = 0,25
$WACC_{20}$ = (0,26 x 80%) + (0,11 x 20%) = 0,23
$WACC_{40}$ = (0,27 x 60%) + (0,12 x 40%) = 0,21
$WACC_{60}$ = (0,28 x 40%) + (0,15 x 60%) = 0,202 ⟹
$WACC_{80}$ = (0,30 x 20%) + (0,18 x 80%) = 0,204
$WACC_{100}$ = (0,32 x 0%) + (0,22 x 100%) = 0,22

∴ lowest WACC = 0,202 ✓2

e) 60% debt ✓2

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